

Proper Linear Sprint Mechanics

Body Part	Motion Summary	Problem	Cause	Drills for Correction
Head	As an athlete initiates acceleration, the head must cast downward. This begins forward movement by slightly placing the head's weight in front of the body's center of gravity. As the athlete approaches maximal running speed, the head is raised to align with the body's vertical axis.	Improper flexion or extension at the head	Fatigue or poor execution of proper mechanics	The athlete maintains proper eye focus ahead or slightly downcast, depending on the specific acceleration or maximal speed drills.
Shoulders	Shoulders should create an exaggerated flexion and extension opposite lower body movement (hip flexion/extension). During acceleration, shoulder extension (driving elbows backward) should cease when the hands pass the buttocks posteriorly. Shoulder flexion should end as the hands rise above forehead level. Proper, aggressive shoulder extension during acceleration and maximal speed aid in proper contralateral movements such as hip and shoulder flexion. Attempts should be made to minimize shoulder abduction and thoracic rotation, keeping the elbows close to the body. This results in the hands approaching but not crossing the body's vertical midline. As the athlete approaches maximal speed, shoulder flexion and extension maintain a high amount of force but movement at the shoulder joint decreases in range of motion. Each shoulder should flex until the hands are even with the nose. Forcible shoulder extension is maintained to move the elbows down and back. The role of arm action shifts from creating propulsive force during acceleration to facilitating the balance of the contralateral hip flexion during maximal speed. During acceleration and maximal speed, shoulders should stay relaxed and swing naturally.	Tight and elevated shoulders during running Movement of the hands across the midline during flexion and extension, causing excessive rotation of the upper body	Arm movement is too forced and not natural Lack of coordination of the contralateral hip and shoulder flexion/extension and overcompensation for poor hip flexion	Standing arm swings; relaxed shoulders while marching or skipping Standing arm swings, A-skips, cycling B-skips, weighted arm swings, and proper technique focus during all drills
Elbows	Elbows should be bent at 90 degrees during shoulder flexion with fingers extended. During shoulder extension, the angle at the elbow may naturally open	Excessive flexion or extension	Poor neural patterning	Standing arm swings, marches, A-skips, cycling B-skips, weighted arm swings,

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slightly but must close to 90 degrees once again during shoulder flexion. Wrists should remain neutral with the fingers extended. This movement pattern is maintained as the athlete approaches maximal speed.

and proper technique focus during all drills

Hips	During acceleration, due to contralateral shoulder extension, hip flexion should reach a point slightly below parallel to the ground. Hip extension should be explosive, yet the posterior movement of the thigh beyond the body's center of gravity should be minimized to less than 20 degrees. As the athlete approaches maximal running speed, hip flexion approaches 90 degrees. Hip extension should remain explosive, yet posterior movement past the body's center of gravity must still be minimized. The pelvis must stay neutral during acceleration and maximal speed.	Inadequate hip flexion Inability to produce adequate force during hip extension (ground push-off) Excessive lumbar extension throughout stride cycle	Poor neural patterning, weak hip flexors, and tight hip extensors Weak posterior chain, tight hip flexors, and inadequate ankle mobility Weak anterior core and glute muscles and tight hip flexors	Flexibility drills for gluteals and hamstrings, resisted knee drives, marches, A-skips, and 1/3/5 wall drill Mobility and flexibility drills for ankle and hip flexors; strength, power, and plyometric drills focusing on the triple extension of the ankle, knee, and hip; 1/3/5 wall drills; and resisted speed drills Core strengthening drills, hip flexor flexibility drills, 1/3/5 wall drills, resisted sprints, and planks
Knees	During the drive phase in acceleration, the knee is fully extended to allow for maximal hip extension with the foot slightly behind the center of mass. The knee then flexes as the hip flexes during the recovery phase, driving the heel directly up toward the buttocks and over the opposite knee. During maximal hip flexion, the knee is flexed, leaving the shin nearly parallel to the ground. As the hip begins to extend again, the knee can swing open to create an angle roughly 90 degrees with the shin before it is rapidly extended again during foot contact. As the athlete approaches maximal speed, the more perpendicular body angle to the ground allows for greater hip flexion and, therefore, higher knee and heel lift.	Knee height is too low during the running stride, not allowing for adequate stride length. Knee adduction during the stance phase of the stride cycle	Inadequate hip flexion during the drive phase of the running stride Poor ankle mobility, weak abductor/external rotators, and tight adductors/internal rotators	Resisted knee drives, marches, A-skips, and resisted sprints Flexibility exercises for gastrocnemius/soleus/Achilles complex, lateral tube walking, single-leg balance drills, and single-leg strength drills
Ankle and Heel	The ankle should remain dorsiflexed to allow for proper foot contact on the ball of the foot during the drive phase. At foot	Inability to maintain dorsiflexion	Improper coaching ("run on the toes"), weak	Mobility and flexibility exercises for gastrocnemius/soleus/

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contact, the angle created between the shin and the foot should be close to 45 degrees, referred to as a *positive shin angle*. This allows the appropriate horizontal force to be applied in the posterior direction. As the foot comes off the ground during recovery, the heel is raised toward the buttocks to *step* over the opposite knee. The height of the heel is related to the height of the knee during hip flexion, so the heel's height begins relatively low during acceleration and increases as maximal speed is approached. Due to overcoming initial inertia during acceleration, the ground contact time of the foot is slightly greater than at maximal speed. As maximal speed is approached, foot contact is still on the ball of the foot but is slightly higher toward the toes. The ankle remains dorsiflexed, but as the foot contact moves slightly to a point directly under or slightly in front of the body's center of gravity, the angle between the foot and shin becomes less acute. As a result, ground contact time for the foot is decreased as maximal speed is achieved.

throughout stride cycle

Foot contact time is too long, resulting in decreased stride rate

tibialis anterior, and tight gastrocnemius/s oleus/Achilles complex

Inability to maintain dorsiflexion, improper backside mechanics, and foot contact too far in front of the center of gravity

Achilles complex, reverse calf raises, A-skips, and 1/3/5 wall drills

Reverse calf raises, A-skips, cycling B-skips, and assisted sprints

Body Angle Concerning the Ground

The body angle should be about 45 degrees to the ground during initial acceleration, allowing maximal force to be created posteriorly. A straight line should be observed from the top of the head, down the spine, and through the extended rear leg. As the athlete approaches maximal speed, the body angle increases to near perpendicular to the ground. This allows the hips to continue to drive the body's center of gravity forward.

Improper body angle at a phase of running (i.e., acceleration versus maximal speed)

Improper coaching ("bend forward" or "stay low" while sprinting) or poor neural patterning

Core strengthening drills, planks, 1/3/5 wall drills, and resisted and assisted speed drills